Whitebox Report (MoneyPoly)

1) Pylint - errors and fixes

1.1 pylint errors (before fixes)

Errors (Before Fixing)

- [code/main.py](code/main.py#L1): C0114: Missing module docstring (missing-module-docstring)
- [code/main.py](code/main.py#L4): C0116: Missing function or method docstring (missing-function-docstring)
- [code/main.py](code/main.py#L11): C0116: Missing function or method docstring (missing-function-docstring)

- [code/moneypoly/bank.py](code/moneypoly/bank.py#L1): C0114: Missing module docstring (missing-module-docstring)
- [code/moneypoly/bank.py](code/moneypoly/bank.py#L2): E0401: Unable to import

'moneypoly.config' (import-error)
- [code/moneypoly/bank.py](code/moneypoly/bank.py#L5): C0115: Missing class docstring (missing-class-docstring)
- [code/moneypoly/bank.py](code/moneypoly/bank.py#L1): W0611: Unused import math (unused-import)

- [code/moneypoly/board.py](code/moneypoly/board.py#L1): C0114: Missing module docstring (missing-module-docstring)
- [code/moneypoly/board.py](code/moneypoly/board.py#L1): E0401: Unable to import 'moneypoly.property' (import-error)
- [code/moneypoly/board.py](code/moneypoly/board.py#L2): E0401: Unable to import 'moneypoly.config' (import-error)
- [code/moneypoly/board.py](code/moneypoly/board.py#L108): C0121: Comparison 'prop.is_mortgaged == True' should be 'prop.is_mortgaged is True' if checking for the singleton value True, or 'prop.is_mortgaged' if testing for truthiness (singleton-comparison)

Line too long (111/100)

Line too long (112/100)

Line too long (111/100)

Line too long (112/100)

Line too long (113/100)

Line too long (113/100)

Line too long (113/100)

Line too long (112/100)

Line too long (111/100)

Line too long (112/100)

Line too long (112/100)

Line too long (120/100)

Line too long (113/100)

Line too long (112/100)

Line too long (112/100)

Line too long (111/100)

Line too long (113/100)

Line too long (112/100)

Line too long (112/100)

- [code/moneypoly/cards.py](code/moneypoly/cards.py#L4):

C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L5):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L6):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L7):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L8):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L9):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L10):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L11):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L12):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L13):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L14):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L15):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L19):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L20):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L21):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L22):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L23):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L24):
C0301: (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L25):
C0301: (line-too-long)

- [code/moneypoly/cards.py](code/moneypoly/cards.py#L26): C0301: Line too long (113/100) (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L27): C0301: Line too long (113/100) (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L28): C0301: Line too long (112/100) (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L29): C0301: Line too long (112/100) (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L30): C0301: Line too long (111/100) (line-too-long)
- [code/moneypoly/cards.py](code/moneypoly/cards.py#L1): C0114: Missing module docstring (missing-module-docstring)

- [code/moneypoly/config.py](code/moneypoly/config.py#L1): C0114: Missing module docstring (missing-module-docstring)

- [code/moneypoly/dice.py](code/moneypoly/dice.py#L1): C0114: Missing module docstring (missing-module-docstring)
- [code/moneypoly/dice.py](code/moneypoly/dice.py#L2): E0401: Unable to import 'moneypoly.config' (import-error)
- [code/moneypoly/dice.py](code/moneypoly/dice.py#L25): W0201: Attribute 'doubles_streak' defined outside **init** (attribute-defined-outside-init)
- [code/moneypoly/dice.py](code/moneypoly/dice.py#L2): W0611: Unused BOARD_SIZE imported from moneypoly.config (unused-import)

- [code/moneypoly/game.py](code/moneypoly/game.py#L453): C0325: Unnecessary parens after 'not' keyword (superfluous-parens)
- [code/moneypoly/game.py](code/moneypoly/game.py#L462): C0325: Unnecessary parens after 'not' keyword (superfluous-parens)
- [code/moneypoly/game.py](code/moneypoly/game.py#L468): C0304: Final newline missing (missing-final-newline)
- [code/moneypoly/game.py](code/moneypoly/game.py#L1): C0114: Missing module docstring (missing-module-docstring)
- [code/moneypoly/game.py](code/moneypoly/game.py#L3): E0401: Unable to import 'moneypoly.config' (import-error)
- [code/moneypoly/game.py](code/moneypoly/game.py#L12): E0401: Unable to import 'moneypoly.player' (import-error)
- [code/moneypoly/game.py](code/moneypoly/game.py#L13): E0401: Unable to import 'moneypoly.board' (import-error)
- [code/moneypoly/game.py](code/moneypoly/game.py#L14): E0401: Unable to import 'moneypoly.bank' (import-error)
- [code/moneypoly/game.py](code/moneypoly/game.py#L15): E0401: Unable to import 'moneypoly.dice' (import-error)
- [code/moneypoly/game.py](code/moneypoly/game.py#L16): E0401: Unable to import 'moneypoly.cards' (import-error)
- [code/moneypoly/game.py](code/moneypoly/game.py#L17): E0401: Unable to import 'moneypoly' (import-error)
- [code/moneypoly/game.py](code/moneypoly/game.py#L20): R0902: Too many instance attributes (9/7) (too-many-instance-attributes)
- [code/moneypoly/game.py](code/moneypoly/game.py#L295): R0912: Too many branches (15/12) (too-many-branches)
- [code/moneypoly/game.py](code/moneypoly/game.py#L381): W1309: Using an f-string that does not have any interpolated variables (f-string-without-interpolation)
- [code/moneypoly/game.py](code/moneypoly/game.py#L402): R1723: Unnecessary "elif" after "break", remove the leading "el" from "elif" (no-else-break)

- [code/moneypoly/game.py](code/moneypoly/game.py#L1): W0611: Unused import os (unused-import)
- [code/moneypoly/game.py](code/moneypoly/game.py#L3): W0611: Unused GO_TO_JAIL_POSITION imported from moneypoly.config (unused-import)

- [code/moneypoly/player.py](code/moneypoly/player.py#L87): C0304: Final newline missing (missing-final-newline)
- [code/moneypoly/player.py](code/moneypoly/player.py#L1): C0114: Missing module docstring (missing-module-docstring)
- [code/moneypoly/player.py](code/moneypoly/player.py#L2): E0401: Unable to import 'moneypoly.config' (import-error)
- [code/moneypoly/player.py](code/moneypoly/player.py#L5): R0902: Too many instance attributes (8/7) (too-many-instance-attributes)
- [code/moneypoly/player.py](code/moneypoly/player.py#L45): W0612: Unused variable 'old_position' (unused-variable)
- [code/moneypoly/player.py](code/moneypoly/player.py#L1): W0611: Unused import sys (unused-import)
- [code/moneypoly/player.py](code/moneypoly/player.py#L58): W0201: Attribute 'in_jail' defined outside **init** (attribute-defined-outside-init)
- [code/moneypoly/player.py](code/moneypoly/player.py#L59): W0201: Attribute 'jail_turns' defined outside **init** (attribute-defined-outside-init)

- [code/moneypoly/property.py](code/moneypoly/property.py#L1): C0114: Missing module docstring (missing-module-docstring)
- [code/moneypoly/property.py](code/moneypoly/property.py#L1): R0902: Too many instance attributes (9/7) (too-many-instance-attributes)
- [code/moneypoly/property.py](code/moneypoly/property.py#L6): R0913: Too many arguments (6/5) (too-many-arguments)
- [code/moneypoly/property.py](code/moneypoly/property.py#L6): R0917: Too many positional arguments (6/5) (too-many-positional-arguments)
- [code/moneypoly/property.py](code/moneypoly/property.py#L48): R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no-else-return)
- [code/moneypoly/property.py](code/moneypoly/property.py#L64): C0115: Missing class docstring (missing-class-docstring)
- [code/moneypoly/property.py](code/moneypoly/property.py#L39): E0202: An attribute defined in property line 17 hides this method (method-hidden)

- [code/moneypoly/ui.py](code/moneypoly/ui.py#L1): C0114: Missing module docstring (missing-module-docstring)
- [code/moneypoly/ui.py](code/moneypoly/ui.py#L69): W0702: No exception type(s) specified (bare-except)

Changes

- code/main.py: Added module docstring + docstrings for get_player_names() and main().
- code/moneypoly/bank.py: Added module docstring + Bank class docstring; removed unused math import; changed import from moneypoly.config to config.
- code/moneypoly/board.py: Added module docstring; changed imports from moneypoly.property/moneypoly.config to property/config; changed prop.is_mortgaged == True to prop.is_mortgaged.
- code/moneypoly/cards.py: Added module docstring; reformatted CHANCE_CARDS and COMMUNITY_CHEST_CARDS entries across multiple lines to

eliminate line-too-long.
- code/moneypoly/config.py: Added a module docstring.
- code/moneypoly/dice.py: Added module docstring; removed unused import of BOARD_SIZE; initialized doubles_streak in **init**.
- code/moneypoly/game.py: Added module docstring; removed unused os + GO_TO_JAIL_POSITION; changed imports from moneypoly.* to local config/player/board/bank/dice/cards and switched to import ui; added pylint disable for too-many-instance-attributes; refactored _apply_card() to a handler map; fixed f-string-without-interpolation in GAME OVER banner; removed no-else-break pattern; removed unnecessary parentheses after not; ensured final newline exists.
- code/moneypoly/player.py: Added module docstring; removed unused sys import; changed import from moneypoly.config to config; removed unused old_position; replaced separate in_jail/jail_turns/get_out_of_jail_cards fields with a jail dict and updated references; addressed missing final newline.
- code/moneypoly/property.py: Added module docstring; added pylint disable for too-many-instance-attributes; changed Property.__init__from multiple positional args to (config, group) to fix too-many-arguments; removed else-after-return in unmortgage(); added **repr** docstring; added PropertyGroup class docstring.
- code/moneypoly/ui.py: Added module docstring; replaced bare except with except ValueError in safe_int_input().

---

2) Errors / logical issues found by white-box tests (Errors 1-18)

Error 1 - Dice was not 6-sided

- Symptom: `Dice.roll()` called `randint(1, 5)` so a value of 6 could never occur.
- Test that found it: `test_dice_roll_uses_six_sided_die`.
- Fix: change randint upper bound from 5 to 6.
- Commit: `1ec0a6a`.

Error 2 - Winner selection was reversed

- Symptom: `Game.find_winner()` returned the minimum net worth player instead of the maximum.
- Test that found it: `test_find_winner_returns_highest_net_worth_player`.
- Fix: use `max(..., key=net_worth)`.
- Commit: `e557dad`.

Error 3 - Passing GO did not award salary

- Symptom: `Player.move()` only paid salary when landing exactly on GO.
- Test that found it: `test_player_move_passing_go_awards_salary`.
- Fix: detect wrap-around and award salary.
- Commit: `483d5a5`.

Error 4 - Player jail state API mismatch

- Symptom: `Game` uses `player.in_jail`, `player.jail_turns`, `player.get_out_of_jail_cards`, but `Player` initially stored jail state differently.
- Test that found it: `test_player_exposes_jail_state_via_attributes`.
- Fix: provide compatibility fields/properties.
- Commit: `c2f2a8f`.

Error 5 - Bank. collect accepted negative values

- Symptom: `Bank.collect(-10)` reduced bank funds even though negatives should be ignored.
- Test that found it: `test_bank_collect_negative_is_ignored`.
- Fix: ignore negative amounts.
- Commit: `1fa7f48`.

Error 6 - Bank.give_loan did not reduce bank funds

- Symptom: Issuing a loan credited the player but didn't reduce the bank's reserves.
- Test that found it: `test_bank_give_loan_reduces_bank_funds_and_credits_player`.
- Fix: route through `Bank.pay_out()`.
- Commit: `2e2421d`.

Error 7 - Board<->Property constructor mismatch

- Symptom: `Board` constructed `Property(...)` in a way that did not match `Property.__init__`.
- Test that found it: `test_board_initializes_without_type_errors`.
- Fix: make `Property.__init__` support Board's construction style.
- Commit: `986cbe4`.

Error 8 - PropertyGroup ownership logic was incorrect

- Symptom: `PropertyGroup.all_owned_by` returned True on partial ownership.
- Test that found it: `test_property_group_all_owned_by_requires_all_properties_owned_by_player`.
- Fix: require `all(...)` and handle empty groups.
- Commit: `70ff69f`.

Error 9 - buy_property rejected exact-funds purchases

- Symptom: `balance == price` was rejected.
- Test that found it: `test_buy_property_allows_exact_funds`.
- Fix: adjust comparison.
- Commit: `471209f`.

Error 10 - pay_rent did not credit the owner

- Symptom: rent was deducted from the tenant but not credited to the owner.
- Test that found it: `test_pay_rent_transfers_to_owner_when_not_mortgaged`.
- Fix: credit owner.
- Commit: `7d9ebb7`.

Error 11 - trade did not credit the seller

- Symptom: buyer cash decreased, but seller cash didn't increase.
- Test that found it: `test_trade_transfers_cash_to_seller_and_property_to_buyer`.
- Fix: credit seller.
- Commit: `6ff27a1`.

Error 12 - mortgage_property did not pay out from bank funds

- Symptom: bank reserves didn't decrease on mortgage payout.
- Test that found it: `test_mortgage_property_pays_out_from_bank_funds`.
- Fix: use `bank.pay_out(...)` and revert on failure.
- Commit: `a5e3dae`.

Error 13 - unmortgage_property cleared mortgage before affordability check

- Symptom: failed unmortgage still cleared the mortgage flag.
- Test that found it: `test_unmortgage_property_does_not_change_state_if_cannot_afford`.
- Fix: check affordability first, only then unmortgage.
- Commit: `171d486`.

Error 14 - Jail fine payment wasn't deducted from the player

- Symptom: choosing to pay the jail fine did not reduce the player's balance (bank collected, but the player wasn't charged).
- Tests that found it:
  - `test_handle_jail_turn_pays_fine_and_moves`
  - `test_handle_jail_turn_declines_card_then_pays_fine`
- Fix: call `player.deduct_money(JAIL_FINE)` before `bank.collect(JAIL_FINE)`.
- Commit: `fe7fb3e`.

Error 15 - Trade accepted negative cash (money minting)

- Symptom: `cash_amount < 0` allowed a trade to increase buyer balance / decrease seller balance (effectively creating money).
- Test that found it: `test_trade_rejects_negative_cash_amount_without_mutation`.
- Fix: reject negative `cash_amount` early without mutating state.
- Commit: `65b43c8`.

Error 16 - collect_from_all / birthday logic didn't charge all donors correctly

- Symptom: transfer-from-all logic did not align with intended behavior in edge cases (eligibility/charging and bankruptcy handling).
- Tests that found it:
  - `test_apply_card_birthday_transfers_from_all_eligible_players`
  - `test_apply_card_collect_from_all_no_eligible_donors_no_transfer`
  - `test_apply_card_transfer_from_all_actions`
- Fix: ensure all required donors are charged per rules and apply bankruptcy checks.
- Commit: `38e3624`.

Error 17 - move_to railroad cards didn't resolve as a property tile

- Symptom: `move_to` card handling treated only `tile == 'property'` as a purchasable tile; `tile == 'railroad'` skipped `_handle_property_tile`.
- Test that found it: `test_apply_card_move_to_railroad_triggers_property_tile_handling`.
- Fix: treat railroad like property in `move_to` card resolution so `_handle_property_tile` runs.
- Commit: `4b226e1`.

Error 18 - CardDeck empty-deck helpers crashed

- Symptom: `cards_remaining()` and `__repr__` crashed on an empty deck.
- Test that found it: `test_card_deck_cards_remaining_on_empty_is_zero_and_repr_does_not_crash`.
- Fix: guard empty-deck behavior and return 0 remaining.
- Commit: `6a870c6`.

---

3) White-box test cases - Why each test case is needed?

test_bank.py

- `test_bank_collect_negative_is_ignored`: Bank collects negative is ignored. (Exposed: Error 5)
- `test_bank_collect_positive_increases_funds`: Bank collect positive increases funds.
- `test_bank_give_loan_reduces_bank_funds_and_credits_player`: Bank give loan, reducing bank funds and credits player. (Exposed: Error 6)
- `test_bank_give_loan_zero_does_nothing`: Bank give loan zero does nothing.
- `test_bank_pay_out_insufficient_raises_value_error`: Bank pay out insufficient raises value error.
- `test_bank_pay_out_negative_returns_zero_and_no_change`: Bank pay out negative returns zero and no change.
- `test_bank_pay_out_zero_returns_zero_and_no_change`: Bank pay out zero returns zero and no change.
- `test_bank_summary_and_repr_branches`: Bank summary and repr branches.

test_cards.py

- `test_card_deck_cards_remaining_counts_down_mod_len`: Card deck cards remaining counts down mod len.
- `test_card_deck_cards_remaining_on_empty_is_zero_and_repr_does_not_crash`: Card deck cards remaining on empty is zero and repr does not crash. (Exposed: Error 18)
- `test_card_deck_draw_cycles_through_cards`: Card deck draw cycles through cards.
- `test_card_deck_draw_on_empty_returns_none`: Card deck draw on empty returns none.
- `test_card_deck_len_and_repr`: Card deck len and repr.
- `test_card_deck_peek_does_not_advance_index`: Card deck peek does not advance index.
- `test_card_deck_peek_on_empty_returns_none`: Card deck peek on empty returns none.
- `test_card_deck_reshuffle_resets_index_and_calls_shuffle`: Card deck reshuffle resets index and calls shuffle.

test_dice.py

- `test_dice_doubles_streak_increments_and_resets`: Dice doubles streak increments and resets.
- `test_dice_reset_and_describe_branches`: Dice reset and describe branches.
- `test_dice_roll_uses_six_sided_die`: Dice roll uses six sided die. (Exposed: Error 1)
- `test_dice_total_is_sum_not_modulo`: Dice total is sum not modulo.

test_game_apply_card.py

- `test_apply_card_birthday_transfers_from_all_eligible_players`: Apply card birthday transfers from all eligible players. (Exposed: Error 16)
- `test_apply_card_collect_from_all_negative_value_is_ignored`: Apply card collect from all negative value is ignored.
- `test_apply_card_collect_from_all_no_eligible_donors_no_transfer`: Apply card collect from all no eligible donors no transfer. (Exposed: Error 16)
- `test_apply_card_collect_increases_balance`: Apply card collect increases balance.
- `test_apply_card_collect_zero_value_no_change`: Apply card collect zero value no change.
- `test_apply_card_invalid_action_no_effect`: Apply card invalid action no effect.
- `test_apply_card_jail_free_increments_card_count`: Apply card jail free increments card count.
- `test_apply_card_jail_sends_player_to_jail`: Apply card jail sends player to jail.
- `test_apply_card_move_to_does_not_pass_go_no_salary`: Apply card move to does not pass go no salary.
- `test_apply_card_move_to_lands_on_property_calls_handle_property_tile`: Apply card move to lands on property calls handle property tile.

- `test_apply_card_move_to_not_property_does_not_call_handle_property_tile`: Apply card move to not property does not call handle property tile.
- `test_apply_card_move_to_passes_go_awards_salary`: Apply card move to passes go awards salary.
- `test_apply_card_move_to_property_with_missing_property_does_not_call_handle_property_tile`: Apply card move to property with missing property does not call handle property tile.
- `test_apply_card_move_to_railroad_triggers_property_tile_handling`: Apply card move to railroad triggers property tile handling. (Exposed: Error 17)
- `test_apply_card_none_returns_no_state_change`: Apply card none returns no state change.
- `test_apply_card_pay_decreases_balance_and_collects`: Apply card pay decreases balance and collects.
- `test_apply_card_pay_negative_value_raises_value_error`: Apply card pay negative value raises value error.

test_game_economy_flows.py

- `test_buy_property_allows_exact_funds`: Buy property allows exact funds. (Exposed: Error 9)
- `test_buy_property_succeeds_when_player_has_more_than_price`: Buy property succeeds when player has more than price.
- `test_mortgage_property_pays_out_from_bank_funds`: Mortgage property pays out from bank funds. (Exposed: Error 12)
- `test_pay_rent_noop_when_mortgaged`: Pay rent noop when mortgaged.
- `test_pay_rent_transfers_to_owner_when_not_mortgaged`: Pay rent transfers to owner when not mortgaged. (Exposed: Error 10)
- `test_trade_rejects_negative_cash_amount_without_mutation`: Trade rejects negative cash amount without mutation. (Exposed: Error 15)
- `test_trade_transfers_cash_to_seller_and_property_to_buyer`: Trade transfers cash to seller and property to buyer. (Exposed: Error 11)
- `test_unmortgage_property_allows_exact_cost_balance`: Unmortgage property allows exact cost balance.
- `test_unmortgage_property_does_not_change_state_if_cannot_afford`: Unmortgage property does not change state if cannot afford. (Exposed: Error 13)
- `test_unmortgage_property_success_charges_player_and_credits_bank`: Unmortgage property success charges player and credits bank.

test_game_find_winner.py

- `test_find_winner_returns_highest_net_worth_player`: Find winner returns highest net worth player. (Exposed: Error 2)
- `test_find_winner_returns_none_when_no_players`: Find winner returns none when no players.

test_game_menus_and_jail.py

- `test_auction_property_accepts_bid_equal_to_minimum_increment`: Auction property accepts bid equal to minimum increment.
- `test_auction_property_covers_pass_too_low_too_high_and_winner`: Auction property covers pass too low too high and winner.
- `test_auction_property_rejects_too_low_bid_to_preserve_winner`: Auction property rejects too low bid to preserve winner.
- `test_handle_jail_turn_declines_card_then_pays_fine`: Handle jail turn declines card then pays fine. (Exposed: Error 14)

- `test_handle_jail_turn_mandatory_release_after_three_turns`: Handle jail turn mandatory release after three turns.
- `test_handle_jail_turn_no_action_increments_turn`: Handle jail turn no action increments turn.
- `test_handle_jail_turn_pays_fine_and_moves`: Handle jail turn pays fine and moves. (Exposed: Error 14)
- `test_handle_jail_turn_uses_get_out_of_jail_card_and_moves`: Handle jail turn uses get out of jail card and moves.
- `test_handle_property_tile_unowned_auction_path`: Handle property tile unowned auction path.
- `test_interactive_menu_exercises_each_choice_branch`: Interactive menu exercises each choice branch.
- `test_interactive_menu_invalid_choice_is_ignored_and_loops`: Interactive menu invalid choice is ignored and loops.
- `test_interactive_menu_loan_amount_non_positive_is_ignored`: Interactive menu loan amount non positive is ignored.
- `test_interactive_menu_loan_amount_zero_continues_to_next_choice`: Interactive menu loan amount zero continues to next choice.
- `test_interactive_menu_positive_loan_amount_calls_bank`: Interactive menu positive loan amount calls bank.
- `test_menu_mortgage_invalid_index_does_not_call_mortgage`: Menu mortgage invalid index does not call mortgage.
- `test_menu_mortgage_no_mortgageable_properties`: Menu mortgage no mortgageable properties.
- `test_menu_mortgage_no_mortgageable_properties_does_not_prompt_for_selection`: Menu mortgage no mortgageable properties does not prompt for selection.
- `test_menu_mortgage_select_valid_property_calls_mortgage`: Menu mortgage select valid property calls mortgage.
- `test_menu_mortgage_selects_second_property_when_available`: Menu mortgage selects second property when available.
- `test_menu_trade_exercises_branch_paths`: Menu trade exercises branch paths.
- `test_menu_trade_invalid_partner_index_returns`: Menu trade invalid partner index returns.
- `test_menu_trade_invalid_property_index_returns`: Menu trade invalid property index returns.
- `test_menu_trade_no_other_players_branch`: Menu trade no other players branch.
- `test_menu_trade_no_properties_branch`: Menu trade no properties branch.
- `test_menu_unmortgage_invalid_index_does_not_call_unmortgage`: Menu unmortgage invalid index does not call unmortgage.
- `test_menu_unmortgage_no_mortgaged_properties`: Menu unmortgage no mortgaged properties.
- `test_menu_unmortgage_selects_second_property_when_available`: Menu unmortgage selects second property when available.
- `test_menu_unmortgage_valid_selection_calls_unmortgage`: Menu unmortgage valid selection calls unmortgage.

test_game_move_and_resolve.py

- `test_move_and_resolve_blank_tile_branch_only_checks_bankruptcy`: Move and resolve blank tile branch only checks bankruptcy.
- `test_move_and_resolve_chance_branch_draws_and_applies_card`: Move and resolve chance branch draws and applies card.
- `test_move_and_resolve_community_chest_branch_draws_and_applies_card`: Move and resolve community chest branch draws and applies card.
- `test_move_and_resolve_free_parking_branch_no_state_change`: Move and resolve free

parking branch no state change.
- `test_move_and_resolve_go_to_jail_branch`: Move and resolve go to jail branch.
- `test_move_and_resolve_income_tax_branch_deducts_and_collects`: Move and resolve income tax branch deducts and collects.
- `test_move_and_resolve_luxury_tax_branch_deducts_and_collects`: Move and resolve luxury tax branch deducts and collects.
- `test_move_and_resolve_property_with_no_property_does_not_call_handle_property_tile`: Move and resolve property with no property does not call handle property tile.
- `test_move_and_resolve_property_with_property_calls_handle_property_tile`: Move and resolve property with property calls handle property tile.
- `test_move_and_resolve_railroad_with_no_property_does_not_call_handle_property_tile`: Move and resolve railroad with no property does not call handle property tile.
- `test_move_and_resolve_railroad_with_property_calls_handle_property_tile`: Move and resolve railroad with property calls handle property tile.

test_game_play_turn.py

- `test_play_turn_banner_turn_number_is_one_indexed`: Play turn banner turn number is one indexed.
- `test_play_turn_doubles_grants_extra_turn_does_not_advance`: Play turn doubles grants extra turn does not advance.
- `test_play_turn_in_jail_branch_handles_jail_and_advances`: Play turn in jail branch handles jail and advances.
- `test_play_turn_non_doubles_advances_turn`: Play turn non doubles advances turn.
- `test_play_turn_three_doubles_sends_player_to_jail_and_advances`: Play turn three doubles sends player to jail and advances.

test_game_remaining_branches.py

- `test_apply_card_none_is_noop`: Apply card none is noop.
- `test_apply_card_transfer_from_all_actions`: Apply card transfer from all actions. (Exposed: Error 16)
- `test_apply_card_unknown_action_is_noop`: Apply card unknown action is noop.
- `test_auction_property_no_bids_path`: Auction property no bids path.
- `test_buy_property_insufficient_funds_returns_false`: Buy property insufficient funds returns false.
- `test_check_bankruptcy_does_not_reset_current_index_when_still_in_range`: Check bankruptcy does not reset current index when still in range.
- `test_check_bankruptcy_eliminates_and_removes_player`: Check bankruptcy eliminates and removes player.
- `test_check_bankruptcy_non_bankrupt_is_noop`: Check bankruptcy non bankrupt is noop.
- `test_check_bankruptcy_player_not_in_list_does_not_remove`: Check bankruptcy player not in list does not remove.
- `test_current_player_and_advance_turn_wraps_and_increments`: Current player and advance turn wraps and increments.
- `test_game_init_constructs_components_and_players`: Game init constructs components and players.
- `test_handle_property_tile_owned_by_other_calls_pay_rent`: Handle property tile owned by other calls pay rent.
- `test_handle_property_tile_owned_by_player_no_rent`: Handle property tile owned by player no rent.
- `test_handle_property_tile_unowned_buy_path_calls_buy_property`: Handle property tile unowned buy path calls buy property.
- `test_handle_property_tile_unowned_skip_path`: Handle property tile unowned skip path.
- `test_mortgage_property_bank_insufficient_reverts_mortgage_flag`: Mortgage property bank insufficient reverts mortgage flag.

- `test_mortgage_property_owner_mismatch_and_already_mortgaged_paths`: Mortgage property owner mismatch and already mortgaged paths.
- `test_pay_rent_owner_none_is_noop`: Pay rent owner none is noop.
- `test_trade_failure_branches`: Trade failure branches.
- `test_unmortgage_property_owner_mismatch_and_not_mortgaged_paths`: Unmortgage property owner mismatch and not mortgaged paths.

test_game_run.py

- `test_run_exits_when_players_reduce_to_one`: Run exits when players reduce to one.
- `test_run_propagates_exception_from_play_turn`: Run propagates exception from play turn.
- `test_run_stops_at_max_turns_and_calls_standings_each_turn`: Run stops at max turns and calls standings each turn.
- `test_run_when_not_running_skips_loop_and_does_not_call_play_turn`: Run when not running skips loop and does not call play turn.
- `test_run_with_no_players_prints_no_winner_message`: Run with no players prints no winner message.
- `test_run_with_one_player_prints_game_over_banner`: Run with one player prints game over banner.

test_player.py

- `test_player_add_money_increases_balance`: Player add money increases balance.
- `test_player_add_money_negative_raises_value_error`: Player add money negative raises value error.
- `test_player_add_property_adds_once_and_remove_property_removes`: Player add property adds once and remove property removes.
- `test_player_deduct_money_decreases_balance`: Player deduct money decreases balance.
- `test_player_deduct_money_negative_raises_value_error`: Player deduct money negative raises value error.
- `test_player_exposes_jail_state_via_attributes`: Player exposes jail state via attributes. (Exposed: Error 4)
- `test_player_go_to_jail_sets_position_and_jail_state`: Player go to jail sets position and jail state.
- `test_player_is_bankrupt_false_when_positive_balance`: Player is bankrupt false when positive balance.
- `test_player_is_bankrupt_true_when_zero_balance`: Player is bankrupt true when zero balance.
- `test_player_move_lands_on_go_awards_salary`: Player move lands on go awards salary.
- `test_player_move_not_on_go_does_not_award_salary`: Player move not on go does not award salary.
- `test_player_move_passing_go_awards_salary`: Player move passing go awards salary. (Exposed: Error 3)
- `test_player_net_worth_equals_balance`: Player net worth equals balance.
- `test_player_remove_property_when_absent_no_change`: Player remove property when absent no change.
- `test_player_repr_contains_name`: Player repr contains name.
- `test_player_status_line_has_jailed_tag_when_jailed`: Player status line has jailed tag when jailed.
- `test_player_status_line_no_jailed_tag_when_not_jailed`: Player status line no jailed tag when not jailed.

test_property_and_board.py

- `test_board_get_tile_type_blank_tile`: Board get tile type blank tile.

- `test_board_get_tile_type_property_tile`: Board get tile type property tile.
- `test_board_get_tile_type_special_tile`: Board get tile type special tile.
- `test_board_initializes_without_type_errors`: Board initializes without type errors. (Exposed: Error 7)
- `test_board_is_purchasable_false_when_mortgaged`: Board is purchasable false when mortgaged.
- `test_board_is_purchasable_false_when_no_property`: Board is purchasable false when no property.
- `test_board_is_purchasable_false_when_owned`: Board is purchasable false when owned.
- `test_board_is_purchasable_true_when_unowned_and_not_mortgaged`: Board is purchasable true when unowned and not mortgaged.
- `test_board_is_special_tile_true_and_false`: Board is special tile true and false.
- `test_board_owned_and_unowned_helpers_and_repr`: Board owned and unowned helpers and repr.
- `test_property_dict_constructor_with_explicit_group_kwarg`: Property dict constructor with explicit group kwarg.
- `test_property_get_rent_base_rent_when_group_not_fully_owned`: Property get rent base rent when group not fully owned.
- `test_property_get_rent_doubles_when_full_group_owned`: Property get rent doubles when full group owned.
- `test_property_get_rent_zero_when_mortgaged`: Property get rent zero when mortgaged.
- `test_property_group_add_property_and_counts_and_repr`: Property group add property and counts and repr.
- `test_property_group_all_owned_by_false_for_none_player_and_empty_group`: Property group all owned by false for none player and empty group.
- `test_property_group_all_owned_by_requires_all_properties_owned_by_player`: Property group all owned by requires all properties owned by player. (Exposed: Error 8)
- `test_property_is_available_false_when_owned_or_mortgaged`: Property is available false when owned or mortgaged.
- `test_property_mortgage_and_unmortgage_branches`: Property mortgage and unmortgage branches.
- `test_property_positional_constructor_registers_group_and_repr`: Property positional constructor registers group and repr.

test_property_based_invariants.py

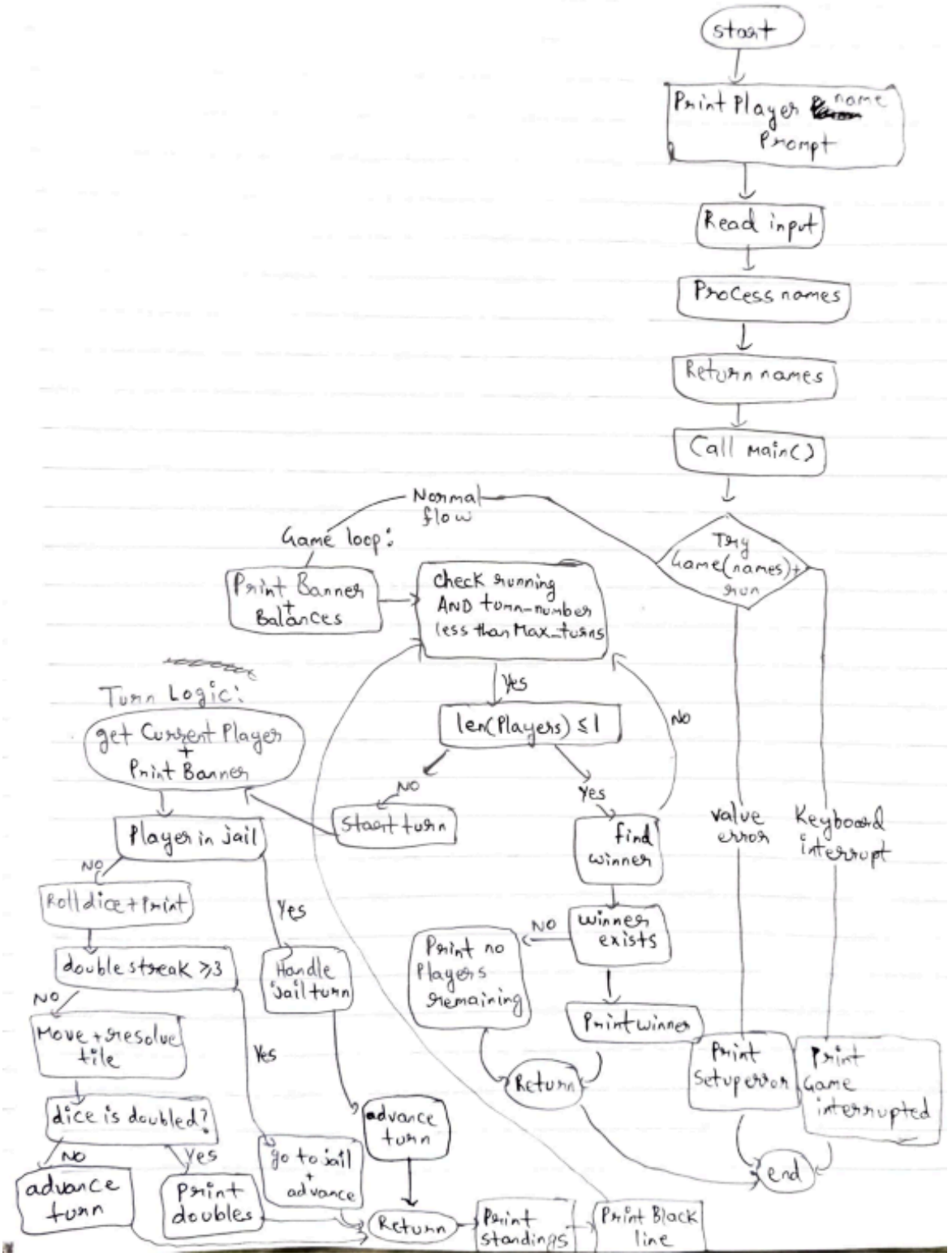
- `test_apply_card_never_raises_for_known_actions`: Apply card never raises for known actions.
- `test_bank_collect_and_pay_out_never_make_funds_negative`: Bank collect and pay out never make funds negative.
- `test_card_deck_draw_cycles_without_exceptions`: Card deck draw cycles without exceptions.
- `test_collect_from_all_is_redistribution_only`: Collect from all is redistribution only.
- `test_simulation_many_turns_no_exceptions`: Simulation many turns no exceptions.
- `test_simulation_seeded_stress_run_no_crash`: Simulation seeded stress run no crash.
- `test_trade_conserves_cash_when_valid`: Trade conserves cash when valid.

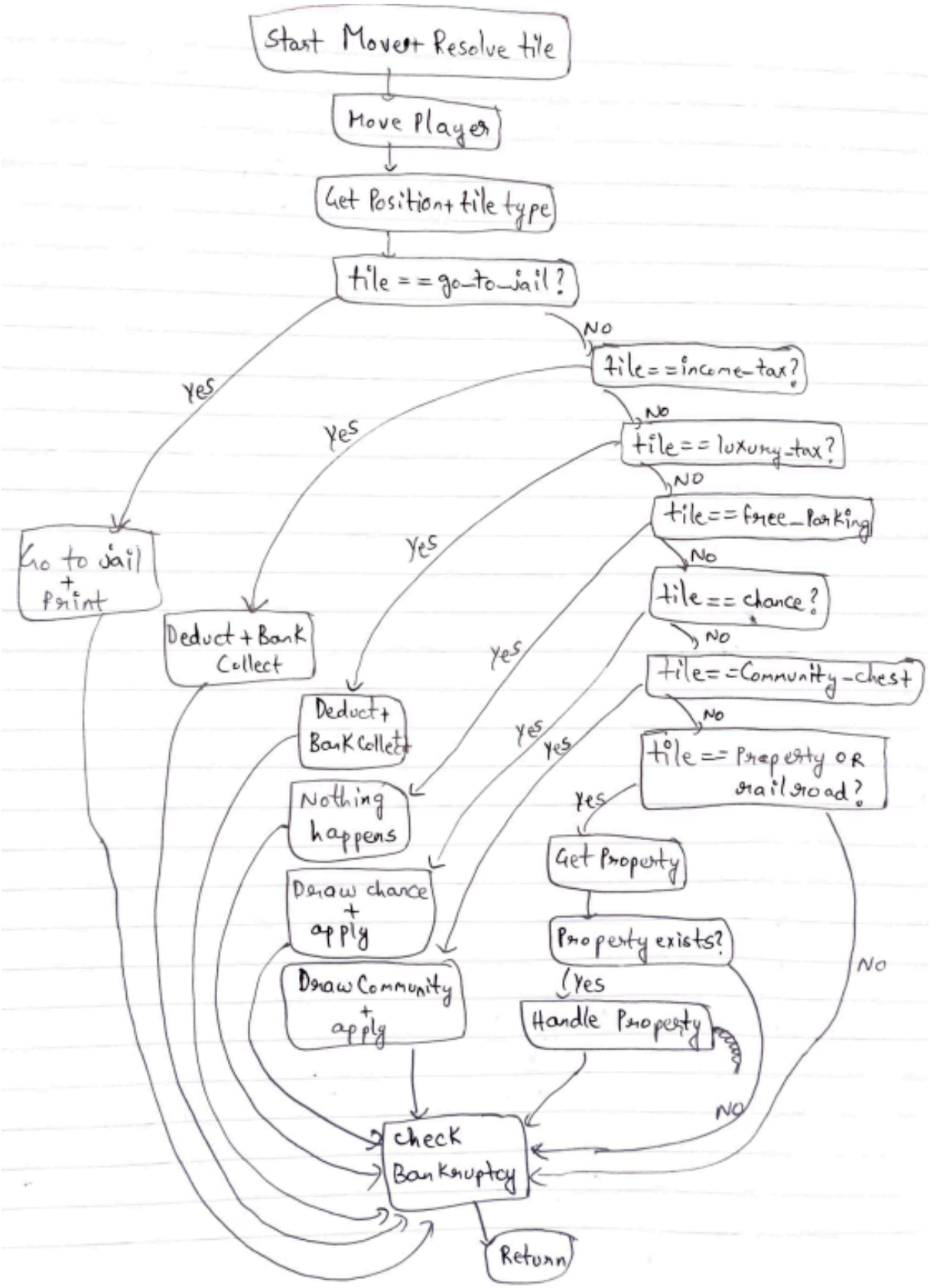
test_ui.py

- `test_confirm_no_variants`: Confirm no variants.
- `test_confirm_unrecognized_input_is_not_yes`: Confirm unrecognized input is not yes.
- `test_confirm_yes_variants`: Confirm yes variants.
- `test_format_currency_handles_zero_and_negative`: Format currency handles zero and negative.
- `test_print_banner_formats_header`: Print banner formats header.

- `test_print_board_ownership_renders_owner_and_mortgage_flag`: Print board ownership renders owner and mortgage flag.
- `test_print_player_card_branches_do_not_crash`: Print player card branches do not crash.
- `test_print_player_card_no_jail_no_properties_branch`: Print player card no jail no properties branch.
- `test_print_standings_sorts_by_net_worth_desc`: Print standings sorts by net worth desc.
- `test_safe_int_input_invalid_returns_default`: Safe int input invalid returns default.
- `test_safe_int_input_valid_number`: Safe int input valid number.

Diagrams:




Start Move + Resolve tile
Move Player
Get Position + tile type
tile == go_to_jail?
Yes
No
Go to jail + Print
tile == income_tax?
Yes
Deduct + Bank Collect
No
tile == luxury_tax?
Yes
Deduct + Bank Collect
No
tile == free_Parking
Yes
Nothing happens
No
tile == chance?
Yes
Draw chance + apply
No
tile == Community_chest
Yes
Draw Community + apply
No
tile == Property OR railroad?
Yes
Get Property
Property exists?
Yes
Handle Property
No
No
check Bankruptcy
Return